PRE-ALGEBRA (8TH) · UNIT 6: CONGRUENCE AND SIMILARITY

Tile Pattern Studio

Design floor tile patterns with transformations, similarity, and angle facts.

CP2

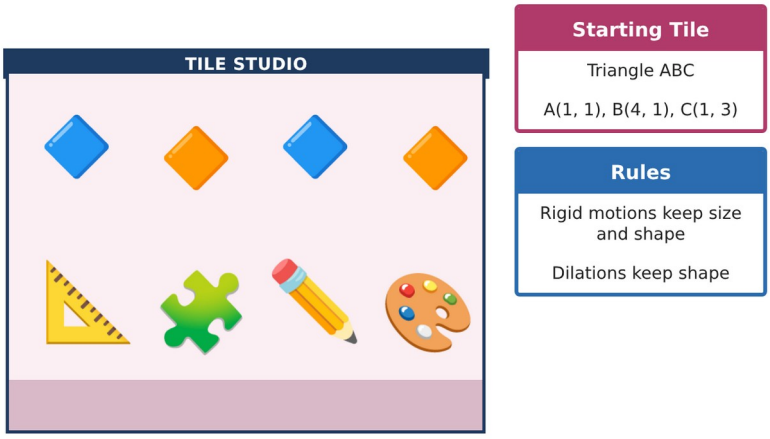
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

A community center is redoing its lobby floor and wants a tile pattern designed by students. The pattern starts with one right-triangle tile that is moved around a grid.

You will use transformations, similarity, and angle relationships to build a pattern that fits perfectly and to measure things you cannot reach.



Learning Goals

- I can translate, reflect, and rotate a figure on a coordinate grid.
- I can dilate a figure and explain why the image is similar.
- I can use angle facts about triangles and parallel lines.

Project Timeline

When	What you will do
Day 1	Slide and Flip
Day 2	Turn and Grow
Day 3	Angle Facts
Final	Tile Design Portfolio + reflection

Part 1: Slide and Flip

Day 1 · Tile ABC: A(1, 1), B(4, 1), C(1, 3).

1. Translate the tile 5 units right and 2 units up. Give the new coordinates.

   Starter: *(x, y) → (x + 5, y + 2)*

2. Reflect the original tile over the y-axis. Give the new coordinates.

3. Are the new tiles congruent to the original? Explain.

Part 2: Turn and Grow

Day 2 · Use the original tile.

4. Rotate the tile 90° counterclockwise about the origin. Use $(x, y) \to (-y, x)$.

5. Dilate the tile by a scale factor of 2 from the origin.

6. Side AB is 3 units long. How long is A′B′ after the dilation? Is the new tile congruent or similar?

Part 3: Angle Facts

Day 3 · Use the triangle angle sum and parallel lines.

7. The tile has a 90° angle and a 34° angle. Find the third angle.

8. Grout lines form parallel lines cut by a transversal. One angle is 115°. Find a corresponding angle and a same-side interior angle.

9. Design a pattern using at least two different transformations of the tile. Describe each move.

Final Product: Tile Design Portfolio

Your final product must include:

- Coordinates for each transformed tile
- A dilated tile with side lengths
- Angle calculations for the pattern
- Your tile pattern on grid paper
- A description of every transformation used

Reflection

R1. Which transformations make a congruent figure? Which make a similar figure?

R2. How do parallel lines help you find angles in the pattern?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Transformations, dilations, similarity, and angle relationships are applied correctly. Goes beyond the minimum with a creative, realistic choice.	Transformations, dilations, similarity, and angle relationships are applied correctly.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Portfolio shows coordinates, rules, measurements, and a clear final design. Polished, organized, and visually engaging.	Portfolio shows coordinates, rules, measurements, and a clear final design. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.